Matrix Computation III: Canonical Forms, Spectral Localization, and Eigenvalue Iteration

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§1.1 Before computing an eigenvalue, decide what can change

A matrix changes when the basis changes. If

$$B = S^{-1}AS,$$

then the entries of B may look nothing like those of A, yet both matrices represent the same linear operator in different coordinates. A useful theory of eigenvalues therefore begins by separating coordinate accidents from invariant structure.

Three questions guide this chapter.

First, what algebraic data survive every change of basis? Canonical forms answer this. Second, before carrying out an expensive computation, where can the eigenvalues possibly lie? Gershgorin's theorem gives a cheap geometric answer. Third, once a spectral region or dominant mode has been identified, how can one actually approximate the corresponding eigenvector? Power and inverse iterations turn localization into computation.

These are different levels of information:

$$\text{canonical structure} \longrightarrow \text{spectral localization} \longrightarrow \text{numerical approximation}.$$

Trying to use one level as a substitute for another causes confusion. Jordan form explains exact structure but is usually a poor numerical algorithm. Gershgorin discs are robust and cheap but rarely exact. Iteration computes selected eigenpairs but only under separation and stability assumptions.

§1.2 Polynomial Matrices and Invariant Factors

The characteristic polynomial

$$\chi_A(\lambda) = \det(\lambda I - A)$$

records the eigenvalues with algebraic multiplicity. It does not, by itself, reveal how many Jordan blocks occur or how large they are. That missing information is already present in the whole polynomial matrix

$$\lambda I - A,$$

not merely in its determinant.

Work over the principal ideal domain $\mathbb{C}[\lambda]$. Row and column operations are allowed when they are invertible over this ring: exchange two rows or columns, add a polynomial

multiple of one to another, or multiply by a nonzero constant. These operations produce polynomial matrices

$$U(\lambda)(\lambda I - A)V(\lambda)$$

with unimodular U, V, meaning their determinants are nonzero constants.

Every polynomial matrix of rank r can be reduced to Smith normal form

$$\operatorname{diag}\big(d_1(\lambda), \dots, d_r(\lambda), 0, \dots, 0\big),$$

where the nonzero polynomials are monic and satisfy

$$d_1 \mid d_2 \mid \cdots \mid d_r.$$

The polynomials d_i are the invariant factors.

If $D_k(\lambda)$ denotes the monic greatest common divisor of all nonzero $k \times k$ minors, then

$$d_1 = D_1, \qquad d_k = \frac{D_k}{D_{k-1}}.$$

The determinant is only the last determinantal divisor:

$$D_n = d_1 d_2 \cdots d_n = \chi_A(\lambda).$$

The smaller minors are what remember how the characteristic polynomial is distributed among cyclic pieces.

§1.3 Invariant factors are a module decomposition

Let $V = \mathbb{C}^n$, and let the indeterminate λ act on vectors by

$$\lambda \cdot v = Av.$$

This turns V into a $\mathbb{C}[\lambda]$-module. The structure theorem for finitely generated modules over a principal ideal domain gives

$$V \cong \bigoplus_i \mathbb{C}[\lambda]/(d_i(\lambda)),$$

where the nonconstant invariant factors are those of $\lambda I - A$.

This description explains why Smith form belongs naturally beside canonical matrix forms. A cyclic summand

$$\mathbb{C}[\lambda]/(d(\lambda))$$

is generated by one vector

$$v, Av, A^2v, \dots,$$

subject to the relation $d(A)v = 0$. In a suitable cyclic basis, the matrix of A on this summand is the companion matrix of d. Taking the direct sum over invariant factors gives the rational canonical form.

The largest invariant factor is the minimal polynomial:

$$m_A(\lambda) = d_n(\lambda),$$

while their product is the characteristic polynomial. Thus

$$m_A \mid \chi_A.$$

The minimal polynomial records the strongest polynomial relation satisfied by the operator and controls every polynomial expression in A, including high powers and matrix functions.

§1.4 From invariant factors to Jordan blocks

Over $\mathbb{C}$, factor each invariant factor into powers of linear terms. The resulting powers

$$(\lambda - \mu)^s$$

are called elementary divisors. Each elementary divisor corresponds to one Jordan block

$$J_s(\mu).$$

For example, suppose the elementary divisors are

$$(\lambda - 2)^2, \qquad (\lambda - 2), \qquad (\lambda - 3).$$

Then the Jordan form is

$$J_2(2) \oplus J_1(2) \oplus J_1(3).$$

The characteristic polynomial is

$$(\lambda - 2)^3(\lambda - 3),$$

but the minimal polynomial is only

$$(\lambda - 2)^2(\lambda - 3).$$

The exponent 2 of $\lambda - 2$ in the minimal polynomial records the largest Jordan block for eigenvalue 2, not the total multiplicity of that eigenvalue.

A concrete polynomial-matrix example makes the mechanism visible. Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Then

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 3 \end{bmatrix}.$$

Because one entry is the unit -1, the gcd of the entries is $D_1 = 1$. Among the 2×2 minors appear both

$$(\lambda - 2)^2 \qquad \text{and} \qquad -(\lambda - 3),$$

whose gcd is 1. Hence $D_2 = 1$. Finally,

$$D_3 = (\lambda - 2)^2(\lambda - 3).$$

The invariant factors are therefore

$$1, \quad 1, \quad (\lambda - 2)^2(\lambda - 3),$$

and the elementary divisors are

$$(\lambda - 2)^2, \qquad (\lambda - 3).$$

They recover the block structure $J_2(2) \oplus J_1(3)$.

§1.5 Jordan Chains and Growth

A Jordan chain for eigenvalue μ is a sequence

$$v_1, v_2, \dots, v_s$$

satisfying

$$(A - \mu I)v_1 = 0,$$
$$(A - \mu I)v_{j+1} = v_j.$$

The first vector is an eigenvector; the others are generalized eigenvectors. In this basis the operator acts by

$$Av_1 = \mu v_1, \qquad Av_{j+1} = \mu v_{j+1} + v_j.$$

Write a Jordan block as

$$J = \mu I + N, \qquad N^s = 0.$$

Then

$$J^k = \sum_{j=0}^{s-1} \binom{k}{j} \mu^{k-j} N^j.$$

The exponential term μ^k is decorated by polynomial factors in k. This is the precise origin of terms such as

$$k\mu^{k-1}, \qquad \binom{k}{2}\mu^{k-2}.$$

For

$$J = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

we obtain

$$J^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Every eigenvalue equals 1, yet powers grow linearly. The Jordan chain carries dynamical information that the list of eigenvalues alone cannot express.

This also explains why Jordan form is mathematically revealing but numerically delicate. A tiny perturbation can split a multiple eigenvalue and destroy a large Jordan block. Computing exact block sizes from floating-point data is therefore ill-conditioned. Stable numerical algorithms usually prefer Schur form,

$$A = QTQ^*,$$

where Q is unitary and T is upper triangular. The diagonal of T gives the eigenvalues while unitary similarity avoids an ill-conditioned eigenvector basis.

§1.6 Gershgorin from Row Sums

Canonical forms describe exact similarity invariants, but they are not a cheap first diagnostic. Gershgorin's theorem locates every eigenvalue using only row sums.

For $A = (a_{ij})$, define

$$R_i = \sum_{j \neq i} |a_{ij}|.$$

The i-th Gershgorin disc is

$$G_i = \{z \in \mathbb{C} : |z - a_{ii}| \le R_i\}.$$

Every eigenvalue lies in

$$\bigcup_i G_i.$$

The proof is a one-line maximum-coordinate argument. If

$$Ax = \lambda x$$

and $|x_i| = \max_j |x_j|$, then

$$(\lambda - a_{ii})x_i = \sum_{j \ne i} a_{ij} x_j.$$

Therefore

$$|\lambda - a_{ii}| \le \sum_{j \ne i} |a_{ij}| \frac{|x_j|}{|x_i|} \le R_i.$$

The chosen coordinate may depend on the eigenvector, so the theorem gives a union of discs rather than one fixed disc.

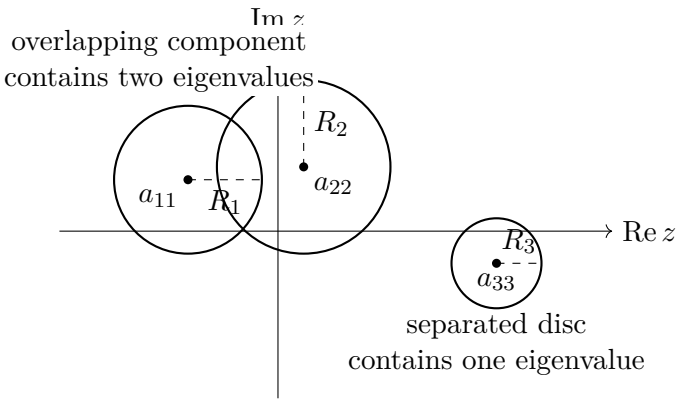


Figure 1.1: A separated component containing exactly m Gershgorin discs contains exactly m eigenvalues, counted with algebraic multiplicity.

§1.7 Counting Eigenvalues with Gershgorin

A stronger form of the theorem says that if a connected component of

$$\bigcup_i G_i$$

contains exactly m discs and is disjoint from the remaining discs, then it contains exactly m eigenvalues, counted with algebraic multiplicity.

The idea is deformation. Consider matrices

$$A(t) = D + t(A - D), \qquad 0 \le t \le 1,$$

where D is the diagonal part of A. At $t = 0$, every eigenvalue is a diagonal entry, and each disc has radius zero. As t increases, the discs expand continuously. If one cluster stays separated from the rest, eigenvalues cannot cross the gap between components. The number inside that component is therefore conserved.

For example, let

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

The first two discs are both the interval-like discs centered at 4 with radius 1, while the third is the point 9. Hence two eigenvalues lie in the component around 4, and one lies at 9. Direct calculation gives

$$\sigma(A) = \{3, 5, 9\},$$

exactly matching the count.

This separated-component principle is particularly useful for proving that a matrix is nonsingular. If every Gershgorin disc avoids the origin, then $0 \notin \sigma(A)$. Strict diagonal dominance is the familiar special case

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|.$$

§1.8 Gershgorin after Diagonal Scaling

Gershgorin discs depend on the chosen basis even though eigenvalues do not. This is not a defect; it is an opportunity. A diagonal similarity

$$B = D^{-1}AD$$

preserves the spectrum while changing off-diagonal magnitudes:

$$b_{ij} = d_i^{-1} a_{ij} d_j.$$

Consider

$$A = \begin{bmatrix} 1 & 100 \\ 0.01 & 2 \end{bmatrix}.$$

The row discs have radii 100 and 0.01, giving a very poor picture. Choose

$$D = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$D^{-1}AD = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Now both radii are 1. The eigenvalues have not moved, but the coordinates have been balanced so that the row equations display their coupling more honestly.

Choosing an optimal diagonal scaling is itself a nontrivial problem. The example nonetheless illustrates a broad numerical principle: poor coordinates can hide useful structure, and similarity transformations can expose it without changing the underlying operator.

§1.9 Power iteration extracts a dominant direction

Suppose A is diagonalizable with eigenpairs (λ_i, v_i), ordered so that

$$|\lambda_1| > |\lambda_2| \geq \cdots.$$

Write the initial vector as

$$x_0 = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n, \qquad c_1 \neq 0.$$

Then

$$A^k x_0 = \lambda_1^k \left(c_1 v_1 + \sum_{j \geq 2} c_j \left(\frac{\lambda_j}{\lambda_1} \right)^k v_j \right).$$

After normalization, the smaller ratios decay, so the direction approaches v_1. The asymptotic rate is governed by

$$\left| \frac{\lambda_2}{\lambda_1} \right|.$$

Take

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

with eigenvalues 3 and 1. Starting from

$$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} v_1 + \frac{1}{2} v_2,$$

we obtain

$$A^k x_0 = \frac{1}{2} \begin{bmatrix} 3^k + 1 \\ 3^k - 1 \end{bmatrix}.$$

The ratio of the two coordinates tends to 1, and the normalized vector approaches the dominant direction $(1, 1)^T$.

A convenient eigenvalue estimate is the Rayleigh quotient

$$\rho(x) = \frac{x^* A x}{x^* x}.$$

For Hermitian A, if x_k approaches a normalized eigenvector, then $\rho(x_k)$ approaches its eigenvalue. The residual

$$r_k = A x_k - \rho(x_k) x_k$$

provides an a posteriori check: an exact eigenvector has zero residual.

§1.10 Power iteration has precise failure modes

The method is not guaranteed merely because a matrix has eigenvalues.

If the initial vector has no component in the dominant eigenspace, the dominant mode never appears. If two eigenvalues tie in modulus, the direction may fail to converge. For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

has two eigenvalues of equal modulus. Starting from a generic vector, the second coordinate changes sign at every step, so the normalized sequence oscillates.

Nonnormality creates another complication. Even when one eigenvalue is asymptotically dominant, an ill-conditioned eigenvector basis can produce large transient mixtures

before the asymptotic regime becomes visible. A small residual is therefore more informative than a visually stable sequence of normalized coordinates.

Finally, normalization prevents overflow but does not repair loss of information. If roundoff repeatedly suppresses a weak component or if the dominant eigenvalue is poorly separated, convergence may stagnate.

§1.11 Inverse iteration turns a target into a dominant mode

Power iteration naturally finds the eigenvalue of largest modulus. To find an eigenvalue near a chosen shift μ, apply power iteration to

$$(A-\mu I)^{-1}.$$

Its eigenvalues are

$$\frac{1}{\lambda_j-\mu}.$$

The eigenvalue of A closest to μ becomes the dominant eigenvalue of the inverse operator.

The practical iteration is not to form the inverse. Instead, solve

$$(A-\mu I)y_{k+1}=x_k$$

and normalize

$$x_{k+1}=\frac{y_{k+1}}{\|y_{k+1}\|}.$$

If the shift stays fixed, factor $A-\mu I$ once and reuse the triangular solves. This makes shifted inverse iteration efficient when several steps are required.

The apparent paradox is that the method becomes stronger as μ approaches an eigenvalue, while the linear system becomes more ill-conditioned. The near singularity is exactly what amplifies the desired eigendirection. Stable pivoting, residual monitoring, and an appropriate stopping criterion are therefore essential.

§1.12 Rayleigh quotient iteration lets the shift move

A natural refinement updates the shift using the current Rayleigh quotient:

$$\mu_k=\frac{x_k^*Ax_k}{x_k^*x_k},$$

then solves

$$(A-\mu_k I)y_{k+1}=x_k$$

and normalizes.

For Hermitian matrices and a sufficiently good initial vector, Rayleigh quotient iteration converges cubically. This remarkable speed comes from two facts: the Rayleigh quotient already approximates the eigenvalue to second order in the angular error, and inverse iteration strongly amplifies the nearest eigendirection.

Take again

$$A=\begin{bmatrix}2&1\\1&2\end{bmatrix}.$$

If

$$x_0=\frac{1}{\sqrt{5}}\begin{bmatrix}2\\1\end{bmatrix},$$

then

$$\mu_0 = x_0^T A x_0 = \frac{14}{5} = 2.8,$$

already close to the dominant eigenvalue 3. Solving with $A - 2.8I$ strongly favors the $(1,1)^T$ direction, and the next Rayleigh quotient is dramatically more accurate.

The method should not be described as universally cubic. Symmetry or normality, a nearby simple eigenvalue, and a sufficiently accurate starting vector are important. For a general nonnormal matrix, convergence can be less regular and the moving linear systems can become difficult.

§1.13 Large matrices lead from vectors to Krylov subspaces

Power iteration uses the one-dimensional space spanned by the newest vector. A more informative approach retains the whole Krylov space

$$\mathcal{K}_k(A, x_0) = \operatorname{span}\{x_0, Ax_0, \ldots, A^{k-1}x_0\}.$$

Instead of asking one vector to represent all spectral information, project A onto this growing subspace.

For Hermitian matrices, the Lanczos process constructs an orthonormal basis of the Krylov space and produces a small tridiagonal matrix. Its eigenvalues, called Ritz values, approximate selected eigenvalues of A. For general matrices, Arnoldi iteration produces an upper Hessenberg projection.

These methods are the large-scale descendants of power iteration. They retain the same polynomial idea—vectors in $\mathcal{K}_k$ have the form $p_{k-1}(A)x_0$—but optimize over many polynomials rather than repeatedly using only $p(t) = t^k$. This is why Krylov eigensolvers can resolve several eigenvalues and converge much faster than plain power iteration.

§1.14 A cubic improvement you can calculate by hand

For the symmetric matrix already used above,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \sigma(A) = \{3, 1\},$$

Rayleigh quotient iteration has an especially transparent calculation. Write a vector in the two eigenvector directions as

$$x \;\propto\; \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \delta \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The number δ measures the relative unwanted eigendirection. Since the two directions are orthogonal, the Rayleigh quotient is

$$\mu = \frac{3 + \delta^2}{1 + \delta^2}.$$

One shifted inverse solve changes the ratio of the two eigencomponents to

$$\delta_{\text{new}} = \delta \frac{3 - \mu}{1 - \mu} = \boxed{-\delta^3}.$$

There is the cubic improvement: in this two-dimensional Hermitian example it is an *exact identity*, provided the shifted linear system is nonsingular.

Take $x_0 = (2,1)^T/\sqrt{5}$, for which $\delta = (2-1)/(2+1) = 1/3$. The original quotient is $14/5$. One step gives

$$\delta_1 = -\frac{1}{27}, \qquad x_1 = \frac{1}{\sqrt{365}}\begin{pmatrix}13\\14\end{pmatrix}, \qquad \mu_1 = \frac{1094}{365} \approx 2.99726.$$

The unwanted component has fallen from $1/3$ to $1/27$ in a single iteration. A second step cubes it again.

There is also a simple certificate for any approximate Hermitian eigenpair. If $\|x\|_2 = 1$, $\mu = x^*Ax$, and $r = Ax - \mu x$, then expansion in an orthonormal eigenbasis gives

$$\|r\|_2^2 = \sum_j |c_j|^2 |\lambda_j - \mu|^2 \;\geq\; \operatorname{dist}(\mu, \sigma(A))^2.$$

Thus some eigenvalue lies within $\|r\|_2$ of the computed quotient. For our x_1, the residual norm is $27/365$, so the interval centered at μ_1 with that radius contains the eigenvalue 3. A rapid iteration and a reliable error check come from the same two-by-two calculation.